Songyu Ye
February 14, 2024
This is what Allen and I talked about on Feb 7.

1 Gelfand Cetlin

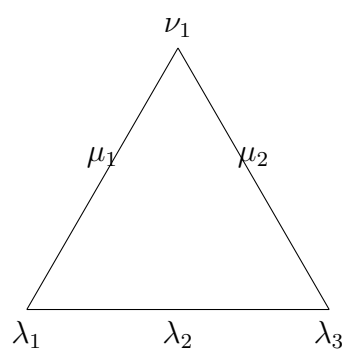
Consider the irreducible representation of $\mathrm{GL}(n)$ with highest weight $\lambda \in \mathbb{Z}^n$ where $\lambda_1 \geq \cdots \geq \lambda_n$. One can consider V_λ as a $\mathrm{GL}(n-1)$ representation and decompose it into irreducible representations of $\mathrm{GL}(n-1)$. Which irreps appear and with what multiplicity? This is the Gelfand Cetlin problem.

The answer is given by the Gelfand Cetlin patterns. In particular,

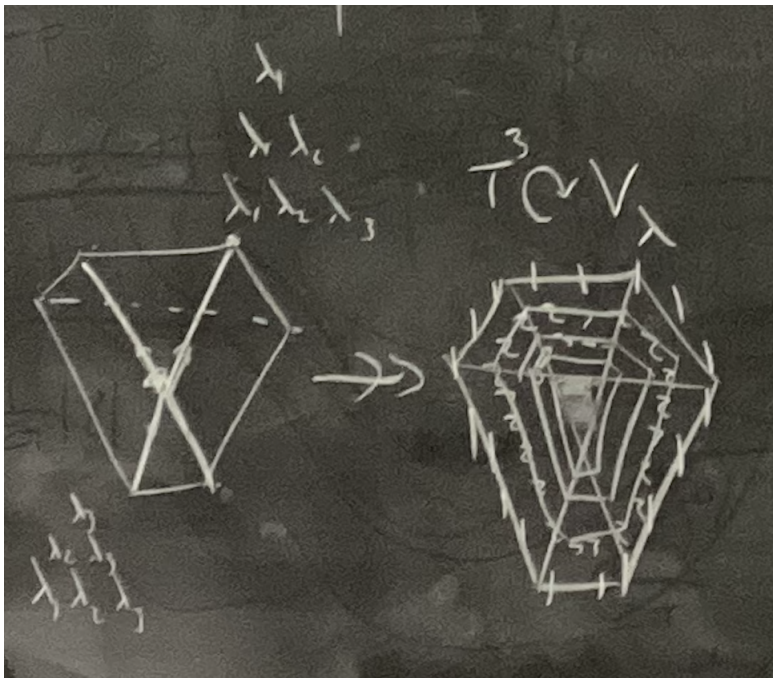
$$V_\lambda = \bigoplus_\mu (V_\mu \otimes \mathbb{C}_{\sum \lambda - \sum \mu})$$

where the sum is over all μ such that $\lambda_1 \geq \mu_1 \geq \lambda_2 \geq \mu_2 \geq \cdots \geq \lambda_n$. Remarkable each such μ shows up and it shows up with multiplicity one.
One can keep going all the way down to $\mathrm{GL}(1)$. The irreps which appear correspond to the different ways of filling out a Gelfand Cetlin pattern.

Example 1.1. *The simplest example is when $n = 3$. In this case we have the following picture:*



We think about all the possible ways to fill out this diagram with the corresponding inequalities. There is a corresponding moment polytope which is the convex hull of the patterns and it looks like



The numbers denote the size of the fiber once you project onto the board. The interesting thing to note about the moment polytope on the left is that it represents a singular toric variety. Remember that the condition that we are looking for is that each vertex has the same number of edges coming out of it. This is not the case for the moment polytope on the left.

The moment polytope has the problem that there is a central vertex with 4 edges, it looks like a pyramid corresponding to the singularity $ad - bc = 0$. Another way of understanding the problem is that there are secret conditions, once you know that you are on the left and right face, then you must be actually at the vertex itself instead of knowing you are at an edge.

There is something in symplectic geometry known as the Thimm trick which we can use in representation theory to extend group actions on vector spaces.

The input data is a group K acting on our vector space V. We can decompose V into isotopic components

$$V = \bigoplus_{\lambda \in \mathfrak{t}_T^*} V_\lambda$$

Then we get an action of another torus T where T acts on each component V_λ by the character λ. This is the Thimm trick. This action of T is not related to the action of K but they commute and so we get an action of $K \times T$ on V. If we repeat this process with $\mathrm{GL}(n)$ acting on V, then we are supposed to end up with an action of $T^{\binom{n}{2}}$ on V. This is the Gelfand Cetlin action?

In algebraic geometry, if we have a group G acting on a ring R, then we can consider R as a G-vector space and again break it up into isotopic components

$$R = \bigoplus_{\lambda \in \mathfrak{t}_T^*} I_\lambda$$

Then one makes the definition

$$R_\lambda = \bigoplus_{\rho} I_{\lambda - \rho}$$

where ρ is sums of the positive roots. In this way we get that $R_\lambda R_\mu \subseteq R_{\lambda+\mu}$ and so we define

$$\operatorname{gr} R = \bigoplus_{\lambda} (R_\lambda / \sum_{\alpha} R_{\lambda - \alpha})$$

Each summand is isomorphic to I_λ. We get an action of $G \times T$.

Example 1.2. *Vinberg asymptotic cone* *Consider the action of* $\mathrm{SL}(2) \times \mathrm{SL}(2)$ *acting on* $\mathrm{SL}(2)$ *by left and right multiplication. We are considering the ring*

$$\begin{aligned} R = \mathbb{C}[\mathrm{SL}(2)] &= \mathbb{C}[a,b,c,d]/(ad - bc - 1) \\ &= \bigoplus_{\lambda} V_\lambda \otimes V_\lambda^* \\ &= \bigoplus_{n} \mathrm{Sym}_n(\mathbb{C}^2) \otimes \mathrm{Sym}_n(\mathbb{C}^2) \end{aligned}$$

the map $\operatorname{gr} : R \to \mathbb{C}[a,b,c,d]/(ad - bc)$ *the degeneration to a singular toric variety. In particular we just forgot the degree* 0 *part.*

In general we want to consider the situation

$$\operatorname{Proj} \bigoplus_n V_{n\lambda} = \mathrm{Fl}(n) \leftarrow \mathfrak{L}_\lambda$$

and we are supposed to apply the same deeneration trick

$$\mathrm{Fl}(n) \to \operatorname{gr}_{n-1} \mathrm{Fl}(n) \to \operatorname{gr}_{n-2} \operatorname{gr}_{n-1} \mathrm{Fl}(n) \to \cdots \to$$

and eventually we are supposed to end up with the toric variety of the Gelfand Cetlin polytope correspodnign to λ.

$\operatorname{Proj} \bigoplus_n V_{n\lambda}$ is supposed to be able to be presented as some polynomial ring, i.e. we are supposed to have a surjection $\mathrm{Sym}(V_\lambda) \twoheadrightarrow \operatorname{Proj} \bigoplus_n V_{n\lambda}$ and in fact it is true that the kernel is generated in degree 2.

The question is we have an action $\mathrm{PGL}(n) \times \mathrm{PGL}(n)$ on $\mathrm{P\bar{G}L}(n) \hookrightarrow \mathrm{Gr}(\dim \mathfrak{g}, \mathfrak{g} \oplus \mathfrak{g})$ and we want to use these tricks to degenerate $\mathrm{P\bar{G}L}(n)$ to a toric variety with the action of $T^{\binom{n+1}{2}} \times T^{\binom{n+1}{2}}$.

The question is what is the polytope.